

Deep State Space Model for Mortality Classification using Electronic Health Records

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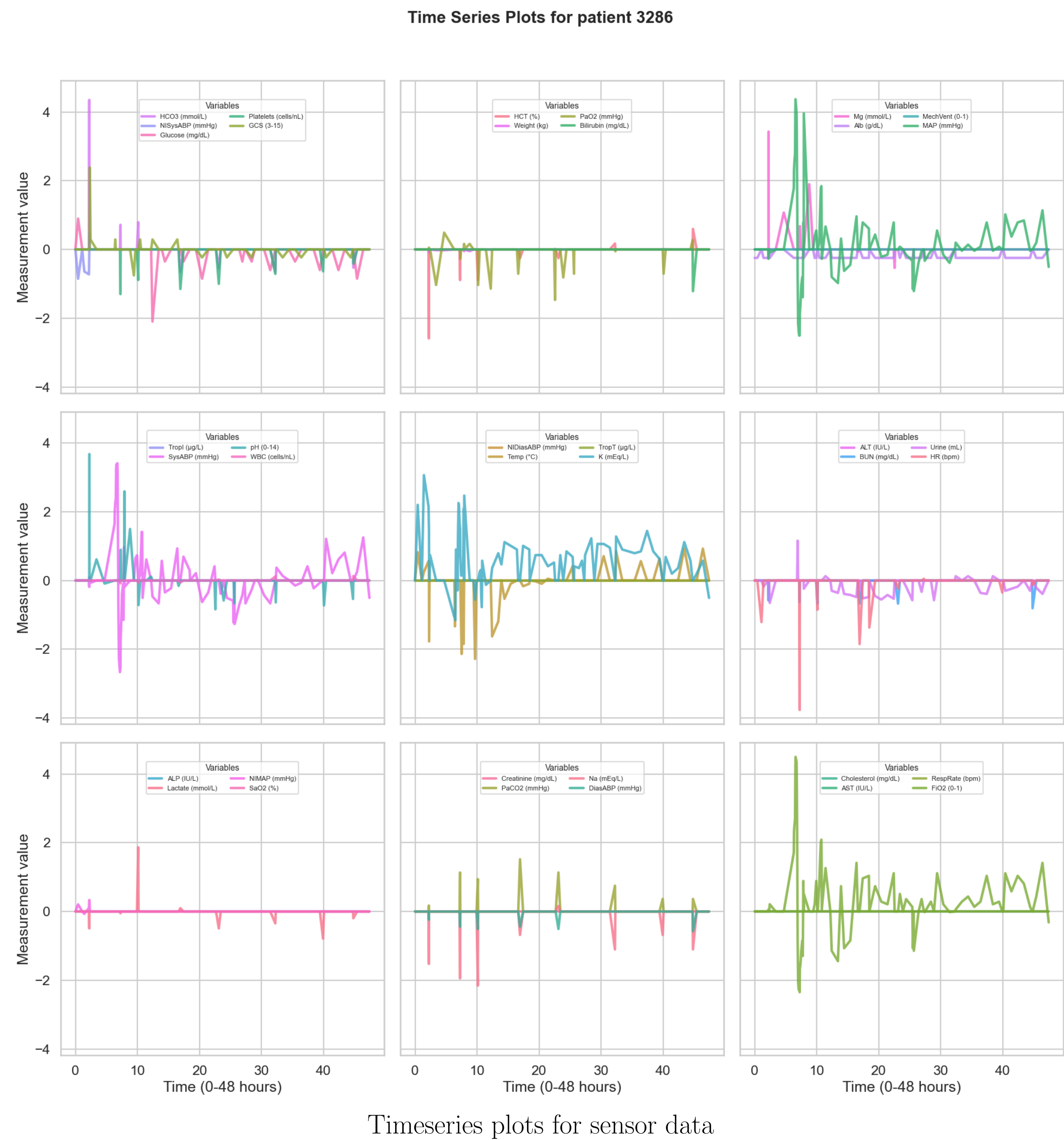
Introduction

The focus of this work was to **reimplement the EHR-Mamba model** for the **PhysioNet2012 dataset**, adapting it to address the unique challenges of this dataset. We investigated if **deep state-space models** are particularly effective for Electronic Health Record (EHR) data, offering advantages over transformers due to their **lower computational complexity** and ability to handle **sparse time-series data**. This poster outlines our approach to **adapting the Mamba model**, the **difficulties** faced during implementation, and the significant results achieved on this dataset.

The PhysioNet2012 Dataset

Description of the Dataset

The PhysioNet/Computing in Cardiology Challenge 2012 [1] dataset comprises records from 12,000 adult ICU stays, collected to develop methods for patient-specific prediction of in-hospital mortality. The dataset includes **37 time-series variables**, and **8 static variables** recorded during the first **48 hours** after ICU admission. Static descriptors, collected upon admission, provide general patient information (Age, Gender, ICU Type), while time-series data consist of multiple observations with associated timestamps indicating the elapsed time since ICU admission. This rich dataset enables the analysis of patient outcome based on early ICU data, facilitating the development of predictive models for mortality. The dataset has been preprocessed [2] [3] and we were able to visualise its components. As visible in the following plots for a random patient, the timeseries are very sparse, with an average sparsity for all physiological measurements of **84.46%**:



Variables in the PhysioNet 2012 Dataset

1. Statics (features) (8 variables): Age, Gender, Height, Weight, Admission Type, ICU Type, Comorbid Conditions, General Patient Status. 1-D array of dimension 8.

2. Time-Series data (37 variables):

In the PhysioNet 2012 dataset, time-series data for each patient is organized with corresponding timestamps and measurements:

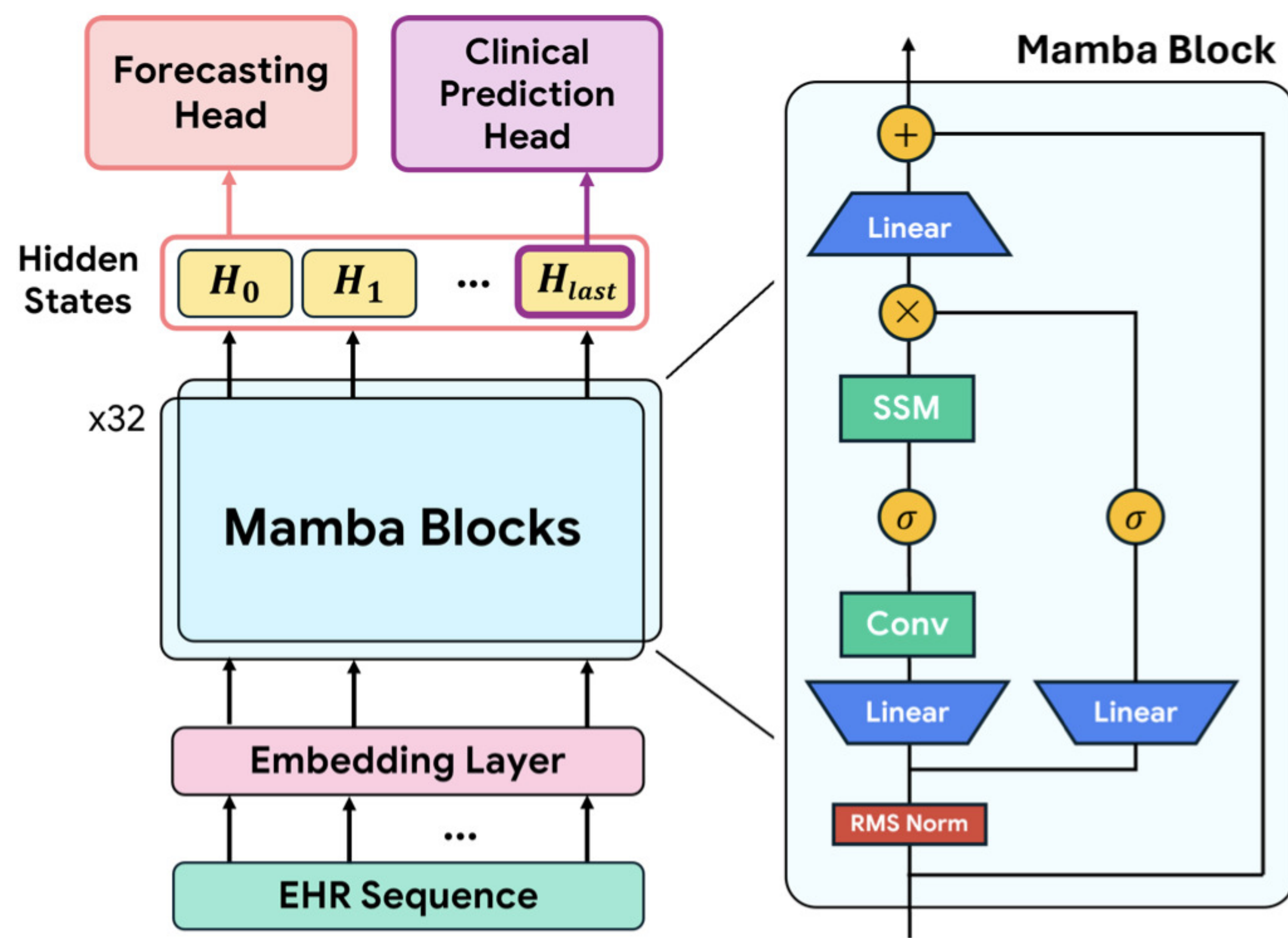
- **ts.times:** This array contains the specific time points at which measurements were recorded during the patient's ICU stay. Each entry represents the elapsed time since ICU admission, typically in hours and minutes. For example, a timestamp of **35.5** indicates that the measurement was taken 35 hours and 30 minutes after admission. This is a 1-D array of **variable length**.

- **ts.values:** This array holds the actual measurements of various physiological variables recorded at the times specified in **ts.times**. Each entry corresponds to a specific variable's value at a given time point. The dimensionality of this feature is a variable number consisting of 37 columns and a variable length of timesteps.

- **ts.indicators** For each time-series variable, a Boolean indicates if a measurement is present at a given time. The dimensionality of this feature is variable and has the **same dimensionality as ts.values**.

3. Labels Binary label: 0 (Survival/Non-critical), 1 (In-hospital Mortality/Critical Outcome). Mortality was observed in **14.23%** of the data, whereas Survival was observed for the remaining **85.77%**.

Overview of the Mamba Block Architecture



Mamba block architecture [4]

The **Mamba architecture** is capable of handling sequences, as **Electronic Health Records (EHRs)**, in order to capture both short-term and long-term patterns efficiently. In contrast to transformers, the Mamba model has **lower computational costs** which is the primary improvement in the field where transformers seemed to be unbeatable for years. For the EHR Sequence we just used the PhysioNet 2012 dataset.

Embedding Layer

- ts.values, ts.indicators, ts.times and static data was used
- Embeddings structure similar to the transformer one which was provided (for now).
- Simplification compared to working with the mimiciV dataset, including:
 - Skipped tokenization since input is already binary (no benefit from tokenization)
 - Multiple visit are not recorded in the PhysioNet dataset

Mamba Blocks

- The input tensor is processed into the first Mamba block and is normalized directly using Root Mean Square (RMS) normalization.
- After normalization, the tensor is expanded through two linear projections.
- The first projection undergoes a convolution, followed by a SiLU activation function (σ). The output is then processed by a discretized State Space Model (SSM), which filters relevant information. On the other side, the second projection directly passes through a SiLU activation function and is then combined with the output from the SSM using a multiplicative gate.

- The combined output is passed through another linear layer.
- Finally, the result is added to the initial input tensor, producing the tensor H_0 .
- The process starts again with H_0 as an input and repeats until H_{last} is defined.

Hidden State Representation and Dual Head Design

- Each block in the model contributes to generating hidden states, which summarize the information of the learned sequence so far. Therefore, the final hidden state H_{last} gathers everything the model has learned about the sequence.
- The model utilizes a dual head design, where each head is responsible for a specific task.
 - With the help of the hidden states, the Forecasting Head can be used to predict what might happen next in the sequence during training. This head helps the model learn the general structure of the data.
 - The Clinical Prediction Head produces a classification that maps the last hidden state to a probability for a specific outcome (in our case mortality).

Baseline Models VS Mamba

Baseline models were provided by our supervisor [5] and they were compared to EHRMamba:

Split #	IPnets		Transformer		GRUD		SEFT		EHRMamba	
Avg.	AUROC	Loss	AUROC	Loss	AUROC	Loss	AUROC	Loss	AUROC	Loss
1	0.8557	0.4019	0.8673	0.4634	0.8737	0.4116	0.8598	0.414	0.8676	0.4879
2	0.8409	0.4019	0.8596	0.4352	0.8630	0.3867	0.8479	0.4366	0.8610	0.4646
3	0.8601	0.5672	0.8674	0.4513	0.8679	0.4208	0.8621	0.3669	0.8673	0.5051
4	0.8498	0.4471	0.8558	0.4104	0.8719	0.3868	0.8475	0.4595	0.8500	0.4650
5	0.8625	0.4171	0.8708	0.4119	0.8677	0.4680	0.8747	0.3342	0.8457	0.5224

Reflection and Potential Next Steps

• Reflection:

- **Tensor Manipulation Challenges:** Significant time was spent on **manipulating tensor sizes**, highlighting the importance of understanding **data structures** in deep learning workflows.
- **GPU Utilization:** Learned how to **train models** efficiently on **GPUs**, including skills in **remote development** and **debugging**.
- **Code Structure Understanding:** Gained a deeper understanding of **class inheritances** and their role in designing **maintainable and modular code**.
- **HPC Optimization:** Effectively leveraged DTU's **HPC** resources, optimizing **training time**.

• Potential Next Steps:

- **Explore Embeddings:** Experiment with **different embeddings** to analyze their impact on model performance.
- **Enhance Forecasting Head:** Train the **forecasting head** even if its results are not directly used, as this could improve **classification accuracy**.
- **Mamba Configuration Testing:** Experiment with **different configurations** of the Mamba model to further optimize results.

References

- [1] Ary L. Goldberger, Luis A. Nunes Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley. PhysioBank, physiotookit, and physionet: Components of a new research resource for complex physiologic signals. 101(23), 2000. URL <https://doi.org/10.1161/01.CIR.101.23.e215>.
- [2] Max Horn, Michael Moor, Christian Bock, Bastian Rieck, and Karsten Borgwardt. Set Functions for Time Series. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*. PMLR, 2020.
- [3] TensorFlow Datasets, a collection of ready-to-use datasets.
- [4] Adibvafa Fallahpour, Mahshid Alinoori, Arash Afkanpour, and Amrit Krishnan. Ehrmamba: Towards generalizable and scalable foundation models for electronic health records, 2024.
- [5] Rachael Marie de Vries. Github repository for model comparison on mortality classification with physionet2012 data (<https://github.com/gsn245/selective-ssm-for-ehr-classification>), 2024.